

Original Research

Seroprevalence and risk factors of *Toxoplasma gondii* infection in women with recurrent fetal loss from the province of Khyber Pakhtunkhwa, Pakistan

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Abstract.

BACKGROUND: In women with a bad obstetric history (BOH), infection is an established cause of recurrent fetal loss. A common infecting agent is the protozoan parasite *Toxoplasma gondii* (*T. gondii*). The aim of this study was to measure the prevalence of toxoplasmosis in women with recurrent fetal loss from the Khyber Pakhtunkhwa province of Pakistan.

METHODS: The study included 360 females aged 16–40 years, of which 180 had a bad obstetric history (study group) and the other 180 had no such history (control group). Blood serum samples were tested for toxoplasma IgM antibodies by Enzyme Linked Immunosorbent Assay and for toxoplasma IgG antibodies using an Immunochromatographic technique.

RESULTS: The overall seroprevalence of toxoplasma infection in study group females was 40.6% and in control group females it was 7.2%. Specifically, IgM prevalence was 12.8% in the study group and 1.1% in the control group. IgG prevalence was 23.9% in the study group and 6.1% in the control group. IgM and IgG combined prevalence was 3.9% in the study group cases. There is a statistically significant association between BOH and seropositivity for *T. gondii* ($p < 0.0001$, Chi square test). Various risk factors associated with *T. gondii* seroprevalence in study and control groups were analyzed.

CONCLUSION: The seroprevalence of toxoplasmosis was significantly higher in women with a bad obstetric history compared to those with no such history. Associated risk factors had no significant effects on the results.

Keywords: Bad obstetric history, infection, recurrent fetal loss, seroprevalence, *Toxoplasma gondii*, toxoplasmosis

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1. Introduction

Not every pregnant woman will enjoy the gift of an alive and healthy baby at the end of a normal pregnancy. Some pregnancies are complicated and have unfortunate fetal outcomes, such as an early trimester abortion, preterm delivery, stillborn

baby or neonatal death and sometimes the baby is born with an anomaly such as hydrocephalus or anencephaly [1–3]. These outcomes can be due to various factors that can affect the course of pregnancy and are devastating experiences to the mother and family [4, 5]. Two out of ten pregnancies are complicated by miscarriage and there is usually a second normal fetal outcome [4]. Sometimes the woman suffers recurrent fetal loss, and these are termed as females with bad obstetric history (BOH) [5]. In these cases, infections are considered an important contributor to undesirable fetal outcome [5, 6] including TORCH (Toxoplasma, Rubella, Cytomegalovirus, Herpes simplex virus II) infections, chlamydia, gonorrhea and syphilis. TORCH infection is an established cause of recurrent fetal loss [7, 8] and toxoplasmosis is suggested to be associated with recurrent fetal loss in many cases [9–11].

Toxoplasmosis is caused by the obligate intracellular protozoan parasite *Toxoplasma gondii* (*T. gondii*). This can be transmitted between its hosts either through vertical transmission (mother to fetus) or by horizontal transmission through ingestion of infected flesh or food or water contaminated with the cysts [12–14]. Toxoplasmosis is self-limiting in humans but can have a serious impact on females who become pregnant whilst harboring the infection and this can lead to various developmental defects in the growing fetus [15, 16]. It may cause stillbirth, preterm delivery or intra uterine fetal death [12–17]. Sometimes a neonate is born with congenital toxoplasmosis, which may lead to various congenital defects in the newborn ranging from asymptomatic disease to serious consequences such as retinal scarring and intracranial calcification hydrocephalus. Such neonates may show signs of delayed growth and psychomotor defects during their early childhood [18–20].

The fetus is more prone to infection if the mother contracts the disease during the course of pregnancy, and the born fetus will manifest the disease in sub-clinical forms ranging up to lethal ocular lesions which can lead to permanent blindness [21]. The risk of congenital toxoplasmosis increases with advancing maternal age, whilst the severity of the disease decreases with advancing gestation [22]. Females infected with *T. gondii* can transmit the infection to the growing fetus as the parasite can easily cross the placental barrier or be transmitted during vaginal delivery [23]. The risk to the fetus depends on the time at which the mother acquires the infection.

Primary infection during the first trimester of pregnancy is associated with the most fatal outcomes for the fetus like spontaneous abortion in the first trimester or it may cause severe manifestations in the developing fetus, including hydrocephaly, microcephaly and chorioretinitis, or it may cause intra uterine fetal growth retardation, premature delivery or intra uterine fetal death [24].

The risk of recurrent fetal loss or babies born with congenital toxoplasmosis is directly related to maternal seropositivity. It is therefore an imperative challenge for health professionals to diagnose and timely treat the disease. It also causes significant social and economic burden for the affected families. The aim of this study was to compare the seroprevalence of *T. gondii* infection in women with recurrent fetal loss to those with no such history in the Khyber Pakhtunkhwa province of Pakistan.

2. Methods

2.1. Study design

This case control study included 360 females of child bearing age from 16 to 40 years. The study was conducted at the outpatient Obstetrics and Gynecology Department of Mardan Medical Complex in Khyber Pakhtunkhwa. Informed written consent was obtained from all participants of the study. Those who were not literate were informed verbally about the study perspectives. A questionnaire was designed to detect socio-demographic plausible causes and risk factors associated with toxoplasmosis and its outcome. Contact with pets and other animals were taken into consideration. The study was approved by the institutional ethical committee of Khyber Medical University, Pakistan (ASRB000039/PB/IBMS/29/04/2013).

2.2. Sampling criteria

Among the 360 females selected for the study, 180 had previous history of two or more abortions, still births, premature deliveries, intrauterine fetal deaths, neonatal deaths, intrauterine growth retardation or a fetus born with hydrocephalus or anencephaly (study group). The other 180 females had no such bad obstetric outcomes during their reproductive history (control group).

Clinical examination was performed to determine anemia, jaundice, clubbing, weight, height and

Table 1
Toxoplasma gondii seropositivity in study and control groups

Group	No. of cases	IgM positive (%)	IgG positive (%)	IgM and IgG positive (%)	Total positive (%)	P
Study	180	23 (12.8%)	43 (23.9%)	7 (3.9%)	73 (40.6%)	<0.0001
Control	180	2 (1.1%)	11 (6.1%)	0	13 (7.2%)	
Total	360	25 (6.9%)	54 (15.0%)	7 (1.9%)	86 (23.9%)	

normal body temperature of the study individuals. Hypertension was excluded by checking blood pressure using a sphygmomanometer - a systolic blood pressure between 100–120 mmHg and a diastolic blood pressure of 70–80 mmHg were taken as normal. Diabetes mellitus was excluded by measuring blood glucose levels. Female with blood glucose levels above normal (80–110 mg/dl) were excluded from the study. Consanguinity was excluded by asking the history. Any female that had recently been vaccinated or used antibiotics were also excluded. ABO and Rh blood grouping were done to exclude Rh incompatibility. Female with uterine anomalies and short cervix were excluded.

2.3. Sample collection

Venous blood samples were collected observing a strict aseptic technique. Approximately 4 ml of blood was collected from a cubital vein. Blood was centrifuged at 4000 rpm for 7 minutes. The clear transparent serum obtained was kept in a sterile gel tube with a cover bearing a serial number and date and stored at -80°C and used for serological analysis.

2.4. Serologic analysis for *Toxoplasma Gondii*

Toxoplasma IgM antibodies were detected in sera by Enzyme Linked Immunosorbent Assay (ELISA) using a commercially available kit from BioCheck (USA). Diluted patient serum sample was added to wells pre-coated with *Toxoplasma* antigen. If present in the patient serum, *T. gondii* specific antibodies bind to the antigen-coated wells. Unbound materials were washed off and HRP-conjugate was added, which binds to the antibody-antigen complex in the tested serum. The excess of HRP-conjugate was washed off and TMB reagent was then added. The enzyme conjugate catalytic reaction was stopped at a specific time interval. The color generated was proportional to the amount of IgM antibody present in the sample. The results were read by a micro-well plate reader (model TC-96, TC TECO diagnostics, USA) set at 450 nm and were compared in parallel with the calibrator and controls. All steps of the assay were performed

according to the manufacturer's instructions. The inter- and intra-assay coefficients of variation were less than 10% across the range of measured results. *Toxoplasma* IgG antibodies were detected by the Immunochromatographic technique (ICT) using kits from Biogenics USA.

The results were analyzed by three independent observers apart from the researchers of the study who were blind to the subject status. Women with positive IgG titers but negative IgM titers were considered latently infected. Women with positive IgG and IgM titers were considered to have a possible recent infection.

2.5. Statistical analysis

Statistical analyses were carried out using SPSS 20 (SPSS Inc., Chicago, IL). The data was expressed as means $\pm$ standard deviation. Chi square test was used to observe association between categorical variables. A *p*-value of less than 0.05 was considered significant.

3. Results

In this study, 180 females had recurrent fetal loss and were labeled as the study group, whilst 180 females of approximately the same age and parity and with no such history were labeled as the control group. The seropositivity of *T. gondii* in study and control groups is summarized in Table 1. Study group females showed a total of 40.6% ($n=73$) positive cases as compared to 7.2% ($n=13$) in the control group. IgM antibodies were positive in 23 (12.8%) study group females as compared to only two (1.1%) in controls. IgG antibodies were positive in 43 (23.9%) of BOH females, whilst control females had 11 (6.1%) positive cases. There were seven (3.88%) cases of BOH females positive for both IgM and IgG, whilst none were positive for both in the control group. There was a statistically significant association between BOH and seropositivity for *T. gondii* ($p < 0.0001$, Chi square test).

Table 2
Relation of IgM status to pet exposure in study and control groups

		IgM status in study group			P
		Negative	Positive	Total	
Pet exposure*	Yes	59	11	70	0.236
	No	98	12	110	
	Total	157 (87.2%)	23 (12.8%)	180 (100%)	
		IgM status in control group			P
		Negative	Positive	Total	
Pet exposure	Yes	61	0	61	0.436
	No	117	2	119	
	Total	178 (98.9%)	2 (1.1%)	180 (100%)	

*Pet exposure was considered as multiple exposures to a pet kept inside homes.

Table 3
Relation of advancing maternal age to IgM and IgG seroprevalence in study and control groups

Age (years)		IgM status			P-value
		Negative	Positive	Total	
Study group	≤25	5 (100%)	0 (0%)	5 (100%)	0.57
	26–30	48 (90.5%)	5 (9.5%)	53 (100%)	
	31–35	57 (83.7%)	11 (16.3%)	68 (100%)	
	≥36	47 (87.0%)	7 (13.0%)	54 (100%)	
	Total	157 (87.3%)	23 (12.7%)	180 (100%)	
Control group	≤25	2 (100%)	0 (0%)	2 (100%)	0.09
	26–30	59 (100%)	0 (0%)	59 (100%)	
	31–35	76 (100%)	0 (0%)	76 (100%)	
	≥36	41 (95.3%)	2 (4.7%)	43 (100%)	
	Total	178 (98.89%)	2 (1.11%)	180 (100%)	
Age (years)		IgG status			P-value
		Negative	Positive	Total	
Study group	≤25	4 (80%)	1 (20%)	5 (100%)	0.95
	26–30	39 (73%)	14 (26.4%)	53 (100%)	
	31–35	53 (77.9%)	15 (22.1%)	68 (100%)	
	≥36	41 (75.9%)	13 (24.1%)	54 (100%)	
	Total	137 (76.1%)	43 (23.9%)	180 (100%)	
Control group	≤25	2 (100%)	0 (0%)	2 (100%)	0.0001
	26–30	59 (100%)	0 (0%)	59 (100%)	
	31–35	74 (97.4%)	2 (2.6%)	76 (100%)	
	≥36	34 (79.1%)	9 (20.9%)	43 (100%)	
	Total	169 (93.9%)	11 (6.1%)	180 (100%)	

3.1. IgM status and its relation to pet exposure in study and control groups

IgM seropositivity and its relation to pet exposure among females of both study and control groups is shown in Table 2. In the study group, pet exposure was found in 70 females (38.8%) and among these 11 (15.7%) were IgM positive. A total of 110 (61.2%) females were found to have no pet exposure. Out of them, 12 (10.9%) were found to be IgM positive showing no statistically significant difference ($p=0.236$, Chi-square test). Out of 180 females in the control group, 61 had a history of pet exposure and all were IgM

negative. Out of 119 females with no pet exposure two were IgM positive showing no statistically significant difference ($p=0.436$, Chi-square test).

3.2. Relation of advancing maternal age to IgM and IgG seroprevalence in study and control groups

The relationship between advancing maternal age and its effects on IgM seroprevalence in study and control groups is shown in Table 3. In study group females, five had an age of ≤25 years and all were IgM negative (100%). There were 53 females aged 26–30 years of which 48 (90.6%) were IgM

Table 4
Types of pregnancy wastage found in BOH study group

Types of pregnancy wastage	Number of cases	Percentage
Abortions	98	54.44%
Perinatal deaths	69	38.33%
Preterm births	8	4.45%
Congenital anomalies	3	1.67%
Intra uterine growth retardation	2	1.11%
Total	180	100.00%

negative and 5 (9.4%) were IgM positive. There were 68 females aged 31–35 years of which 57 (83.8%) were IgM negative and 11 (16.2%) were IgM positive. There were 54 females aged ≥ 36 years of which 47 (87%) were IgM negative and 7 (13%) were IgM positive. Thus, in the study group females, IgM seroprevalence showed increasing trend with advancing maternal age. The overall IgM seroprevalence among study group females was 12.7% (the difference was not statistically significant $p > 0.05$, chi-square test). In control group all age females were found to be IgM negative except the age ≥ 36 , there were 41 females, among them 2 (4.7%) were IgM positive.

The relationship between advancing maternal age and its effects on IgG seroprevalence in study and control groups is also shown in Table 3. In the study group, five females were aged ≤ 25 years of which one (20%) was IgG positive and four (80%) were IgG negative. Out of 53 females aged 26–30 years 14 (26.4%) were IgG positive. Out of 68 females aged 31–35 years 15 (22.1%) were IgG positive. Out of 54 females aged ≥ 36 years 13 (24.1%) were IgG positive. Out of 180 females in the study group a total of 43 (23.9%) were IgG positive. However, in all the age groups the difference was not statistically significant $p > 0.05$ (chi-square test). In the control group, the two females aged ≤ 25 years were IgG negative. Out of 59 females aged 26–30 years all were IgG negative. Out of 74 females aged 31–35 years, two (2.6%) were IgG positive. Out of 36 females aged ≥ 36 years nine (20.9%) were IgG positive.

3.3. Types of pregnancy wastage found in the study group

The type of pregnancy wastage found in the study group is shown in Table 4. Out of 180 BOH study group females, it was found that abortions were the most frequent type of pregnancy wastage accounting for 98 cases (54.4%), followed by 69 cases of perinatal deaths (38.3%), eight preterm births (4.45%), three congenital anomalies (1.7%) and two cases of intrauterine growth retardation (1.11%).

4. Discussion

In the present study, the prevalence rate of Toxoplasma infection in females with recurrent pregnancy loss was found to be 40.6%, whilst in females with no history of recurrent pregnancy loss the rate was found to be 7.2% (Table 1). Our study shows a high prevalence rate in females with recurrent pregnancy loss, than that from a study conducted in the district of Kohat in Khyber Pukhtunkhwa, which demonstrated an overall 14.4% toxoplasma seroprevalence in pregnant women irrespective of their past history of pregnancy loss [23, 24]. Toxoplasmosis is a zoonotic disease with worldwide distribution among various communities [25]. The prevalence rate varies around the world due to wide differences in the nature of food consumed, variations in the method of cooking, personal hygiene and the approach to handling of animals [26]. Infected mothers and babies born to them are more prone to various congenital infections and other complications which may end the pregnancy in early abortions, preterm births and perinatal mortalities [27]. These drastic effects in pregnant females may be due to a low maternal immunity state and prevailing anemia during pregnancy. The trend of toxoplasma prevalence in neighboring Asian countries is variable. According to a study conducted in New Delhi, India a high prevalence rate of 45% among pregnant females was found [25]. Another study conducted in Andhra Pradesh, India showed 49.5% prevalence among women with bad obstetric history and the control group had a lower rate of 12.4% [26]. In different provinces of China, prevalence in the general population varies from 12.3% to 25.4% [22, 27]. Studies conducted in Japan [13–27] and Thailand reported prevalence rates in pregnant females of 10.3% and 28.3%, respectively [28].

In our study, BOH females showed 12.8% toxoplasma IgM seropositivity as compared to 1.1% among the control group, whilst toxoplasma IgG antibodies were positive in 23.9% of BOH females and in 6.1% of controls (Table 1). These results are comparable to a study from Andhra Pradesh, India, which reported 27.6% IgG seropositivity in BOH females and 9.5% in controls [26].

Because toxoplasmosis is a zoonotic disease, pets are considered as a possible threat for acquiring the infection as demonstrated in various studies [23, 26]. However, our results failed to reach the statistical significance and we found no significant difference between females of the study and control groups regarding IgM seropositivity and pet exposure

(Table 2). These observations could reflect cultural, social and religious aspects of our society because in a society like Pakistan, people usually try to avoid direct contact with pets especially cats and dogs. Whilst in countries like China, Japan, Thailand, and other western countries, pets are usually kept indoors and direct handling by owners is quite frequent, thus providing an increased window of getting infected with the parasite. The association of pets, especially cats, dogs and cattle with human toxoplasmosis is usually difficult to assess by simple epidemiological studies. This is because the contaminated soil, water and food are the risk and not the pet itself, as the cysts are not found on the fur or body of the pet, but they are excreted in the feces to the external environment.

In our study, there was a general trend of higher IgM seroprevalance in females with advancing maternal age. In BOH females, a maximum of 16.2% IgM seropositivity was found for the age group 31–35 years, whilst those aged ≤ 25 years had no IgM seropositivity. In control group females, IgM and IgG seropositivity was only detected in age groups of 31–35 years and ≥ 36 years. This observation agrees with studies conducted in other parts of the world. In Asam, India, an increased prevalence of toxoplasma infection was found in mothers of older age group. The study by Borkakoty et al., reported a high prevalence rate of 60.7% with advancing maternal age [22]. A study conducted in Brazil demonstrated a gradually increasing trend of toxoplasmosis with increasing maternal age, the highest in an age group of ≥ 30 years [21]. A high prevalence rate can be due various other associated factors that affect maternal health and immunity in females of older age groups. Associated factors include increased anemia with recurrent pregnancies and abortions, lower immunity status with advancing maternal age and more frequent chances of exposure to the parasite from contact with environmental conditions.

This study report a significant difference in terms of recurrent pregnancy loss in the cases relative to control group ($p = 0.0001$, Chi square test). The types of pregnancy loss in the study group had abortion as the most frequent (54.4%) followed by perinatal deaths (38.3%), including stillbirths and early neonatal deaths (Table 4). Preterm birth accounted for 4.5% of fetal loss, congenital anomalies for 1.67% and Intra Uterine Growth Retardation (IUGR) for 1.1% of cases (Table 4). These observations are comparable to those from an Indian study where they reported 51.9% cases of abortions as the most frequent type of pregnancy wastage, followed by still

births (36.5%), preterm deliveries (7.7%), congenital anomalies (1.9%) and early neonatal deaths (1.93%) [25, 26]. Another study from India also reported abortion (27.2%) as the most common type of pregnancy loss in females screened for TORCH infection [27–29]. Among the limitation of the study is the sole screening of females for toxoplasma infection. We were unable to include pets as possible factor for toxoplasma infection. The screening of pets would have further strengthened the study.

We recommend further studies in other parts of the country to find the seroprevalance of toxoplasmosis in females of child bearing age in order to reduce recurrent pregnancy loss and associated congenital toxoplasmosis in newborns born to infected mothers. Routine serological investigation of all pregnant females with or without bad obstetric history should be carried out for both IgM and IgG toxoplasma antibodies. Those females who are IgM as well as IgG positive are a challenge because it is difficult to distinguish between recent and past infection. They must be followed and reassessed after 4–6 weeks for seroconversion.

5. Conclusion

In Khyber Pakhtunkhwa province of Pakistan, the seroprevalance of toxoplasmosis was significantly higher in women with a bad obstetric history compared to those with no such history. Associated risk factors had no significant effects on the results. Because of the established link between toxoplasmosis and recurrent fetal loss, studies like this provide important information for health professionals to diagnose, monitor and properly treat the disease. Such studies also allow the prevalence of the disease to be compared in different regions and in different parts of the word, thus contributing to the global picture of public health.

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